Could the Fighter Pilot of the Future Still be Human?



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Table of Contents

1. Introduction *3*

2. Defining Autonomy and Pilot Replacement *3*

3. AI as the Aircraft's Controller *4*

4. AI as the Aircraft's Tactical Authority *5*

5. Analytical Framework *6*

6. Conclusion *10*

7. Evaluation *11*

8. Bibliography *13*

1. Introduction

The prospect of AI taking control of the military cockpit at first raises the question: 'can artificial intelligence systems do everything the human pilot can?'. However, to meaningfully provide insight to this, we must first define what a fighter pilot is. Most tasks carried out by military aviators fall under two categories: controlling the physical aircraft and tactical decision making. The former entails flying, navigating and manoeuvering the jet, while the latter encompasses reacting to threats, initiating combat and weapons handling. Overlooking this distinction could mislead us on what the fighter pilot of the future looks like.

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2. Defining Autonomy and Pilot Replacement

As a precursor to artificial intelligence, many systems already use automation [4]. This is the process of carrying out pre-defined tasks without human input, and is currently used in routine, repetitive tasks such as landing and take-off protocols. AI is extending the limited scope of this technology by introducing the ability to select and carry out an action with minimal human intervention, a characteristic called autonomy.

Replacement of humans within the cockpit is not a binary process. There are two useful frameworks that are used to facilitate human-AI teaming: human-on-the-loop (HotL), or human-in-the-loop (HitL) [10]. HotL describes a situation where the system takes full advantage of machine autonomy to perform tasks without having to wait for human input. It is an example of a model where a human is supervising an autonomous system. On the other hand, human-in-the-loop systems are those where the human crew intervenes to make decisions at important points within the operation of the aircraft.

In this paper, I will first discuss AI as the aircraft's controller and then AI as the aircraft's tactical decision maker. Using this information, I will judge replaceability by examining aspects of a pilot's job through the following criteria to test my hypothesis that AI will first replace humans as the aircraft controller before replacing humans as the tactical authority.

1. **Predictability**: how formulaic is the task?
2. **Consequences of Error**: what impacts do mistakes have?
3. **Human Input**: how much human input is there currently, and how difficult will it be to completely remove the human from the loop?

4. **Judgement**: how much judgement and analysis of the current scenario is needed?
5. **Outcome Verifiability**: is there a fixed correct result?

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3. AI as the Aircraft's Controller

While autopilot has an established role in civilian aviation, military aviation adds a layer of complexity due to its exposure to rapidly changing environments. However, programs and systems are currently in development to increase the level of automation within fighter aircraft.

One such example is the Auto Ground Collision Avoidance System (Auto GCAS), currently operational on more than 600 U.S. Air Force F-16s [2]. The aim of this technology is to reduce Controlled Flight into Terrain incidents (CFIT), and it automatically becomes operational if the pilot is unresponsive to a potential crash alert [2]. This highlights automated systems' capacity to carefully maneuver a plane in high-risk scenarios. However, this does not prove that such technology can replace humans as the overall aircraft controller, since Auto GCAS is designed for a very specific use case. The system's success in this particular task, therefore does not translate to the AI being able to operate the fighter jet under the full range of conditions.

The United States' Defense Advanced Research Projects Agency (DARPA) has developed and completed a program called ACE (Air Combat Evolution) [6]: a program developed with the aim of autonomously controlling fighter aircraft. ACE demonstrates AI's capability to maneuver an aircraft successfully in more dynamic situations than Auto GCAS can. Another such example is the Loyal Wingman Program [3][1], where an unmanned aircraft serves as a secondary to a leading manned aircraft, detecting threats and engaging with them. These examples help strengthen the argument that intelligent systems could replace humans as the physical controller of fighter jets in the future. However, this technology still needs to be scaled to more 'complex' [6] and 'multi-aircraft' [6] scenarios. This limitation is significant as replacement of the human as a fighter pilot requires the ability to perform in more scenarios than controlled trials [5].

The evidence therefore suggests that a replacement of the human aircraft controller requires adaptability, skill and reliability. At the moment, current evidence does not demonstrate AI's ability to do so across the full range of operating conditions. However, I believe that as the technology continues to advance and

programs such as ACE take a more prominent role in military aviation, the human pilot's role as the aircraft controller could be replaced by intelligent systems.

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4. AI as the Aircraft's Tactical Authority

One process in decision making is anomaly detection, which can be carried out using Hierarchical Cluster Analysis or the Density Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm [7]. Both these systems flag observations that are sufficiently dissimilar from clusters representing normal functionality. However, we must consider the possibility that these apparent anomalous conditions are in fact safe, or that seemingly normal clusters represent genuine failures. This could result in the production of false positives and negatives, misleading pilots and potentially resulting in safe conditions being unintentionally avoided. As useful as these anomaly detection algorithms are at reducing pilot work-load, humans are still needed to review the results produced from these algorithms before they can reliably inform tactical decisions.

Another part of decision making is target recognition. One such method currently in development [8] is the You Only Look Once (YOLO) algorithm. This system uses computer vision to process an image of the aircraft's surroundings and can then identify potential targets rapidly. However, as with any AI system, YOLO is sensitive to the quality of training data and the bias within it. For example, the model could struggle to identify specific types of aircraft or objects. Moreover, since such systems are still being developed, a key restriction is their inability to reliably identify smaller objects. But does imperfect target recognition automatically prevent AI from taking over as the decision maker when humans are capable of the same mistakes? This prompts the discussion that humans, regardless of experience or skill, will always be vulnerable to misjudgement or human error. As AI training data continues to develop and become more detailed, the aviation sector could potentially reach a stage where intelligent systems are able to carry out target identification more reliably than humans can. More importantly for the future relationship between humans and AI, however, is the shift from analysing a scenario to taking action.

Current military aviation is seeing the introduction of Reinforcement Learning systems as part of decision support within the cockpit. An example is a State Value Function [7] which estimates expected future rewards for different states, helping the pilot decide what manoeuvre to select. However, if the reward is poorly specified, the algorithm could end up favouring negative scenarios.

Another key issue is sim-to-real transfer. This is the shift from testing an algorithm with a computer simulation to implementing it in an operational fighter jet. This creates discrepancies which impact the functionality of the algorithms and their ability to effectively produce recommendations in real life scenarios. These limitations illustrate that while AI can move beyond simply recognising an issue to selecting possible actions, it is difficult to rely on these systems when performing in scenarios different to simulated conditions.

So far, we have discovered that AI can detect and identify threats as well as generate decisions based on this information. So what is stopping intelligent systems taking over as the tactical authority? A key point to remember is that being capable of selecting an action is not the same as being given the control to execute it, especially since AI lacks the ability to exercise judgement or make moral decisions at the moment [9]. Issues with object mis-identification carry more weight as weapons could be fired at the wrong target. As such, critical decisions still require human intervention.

In summary, technical capability alone is not sufficient to support a full transfer of tactical authority.

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5. Analytical Framework

From the discussion in Sections 3 and 4, I have made my own framework to assess the replaceability of humans with AI in the cockpit of a fighter aircraft. **Table 1** on the following page compares 5 pilot functions against the criteria outlined in Section 2. I chose these pilot functions as they are a mix of both controlling the aircraft and tactical decision making. It is important to note that my model qualitatively compares different pilot functions against one another.

Table 1 - Pilot Role Analysis Matrix:

Pilot Function	Predictability	Consequences of Error	Human Input	Judgement	Outcome Verifiability
Maintain flight	Highly repeatable; defined parameters (altitude, speed); overall aim is constant	Ground collisions, mid-air collisions; endangering lives; humans and AI fail in different ways; humans impacted by cognitive overload or distraction [13]	Significantly reduced; automated flight-control systems in use [11]; less human input than tactical decision making	Standard procedures for flight; sudden weather changes could require judgement; situational awareness is key	Relatively easier; data such as speed, altitude, flight path can be measured objectively; strategic decisions more subjective
Navigation	Predictable due to planned route [12]; overall flight must adhere to objectives and constraints	Mistakes could be corrected before serious harm; potential to be quite impactful; aircraft could venture into unsafe airspace	Humans already heavily assisted by automated systems; comparatively easier to fully remove human from the loop	Set instructions; judgement may be needed in crowded airspace; level of judgement considerably lower than tactical decision making	Fixed flight path; outcome judged objectively; position of aircraft compared with set waypoints; comparatively easier to verify outcome
Threat assessment	Highly unpredictable; each threat behaves differently; process by which threats are assessed could be repeatable; AI could introduce more consistency for each assessment	False positives / negatives; could put pilot in an underestimated threat; knock-on effects - could impact the tactical decisions made and corresponding actions	Humans still needed to review results; not fully accurate algorithms yet	Lots of judgement needed; significance of threat dependent on behaviour, location, surrounding environment; different models may view same threat differently; AI must respond to changing scenarios	Uncertainty and multiple reasonable conclusions makes one definitive answer challenging; harder to determine if correct decision was made

Pilot Function	Predictability	Consequences of Error	Human Input	Judgement	Outcome Verifiability
Tactical decision making	Extremely unpredictable; each situation will have multiple possible actions; decisions made in the moment; very hard to predict exact conditions pilot will face	High risk; pilot engaging with the wrong aircraft; ignoring a genuine threat; harder to correct than areas such as navigation; could have immediate impacts; wrong decisions impact trustworthiness of system	Humans retain final control over any algorithms - make final decision over what action to pursue; immense reliability needed which current AI systems do not yet have	Extremely important; actions taken will be different depending on scenario; presence of civilian aircraft impacts decisions made	No one correct 'answer'; different people / AI systems will have different opinions; compared to navigation and control, much harder to determine the status of the AI's decision
Execution of decision	More predictable than decision making; specific objectives for each action; split second changes in combat can alter the suitability of an action	Potentially very severe; mistakes could be irreversible; high reliability needed	Low when action is being executed; humans still retain control over whether the action is actually carried out	Once action has been decided, less judgement is needed; unforeseen circumstances could require on-the-spot analysis [14]	Comparatively easier than tactical decision making to determine whether task was carried out correctly

From this information, we can then sort the five pilot roles based on their relative levels of predictability, consequences of error, human input, judgement and outcome verifiability. In **Table 2** below, a ranking of 1 corresponds to the highest level, while a ranking of 5 represents the lowest level.

For my model, the pilot function that is most likely to be fully automated will be the one that has the highest predictability and outcome verifiability, combined with the lowest consequence of error, human input and judgement.

Table 2 - Comparative Ranking of Pilot Roles:

Rank	Predictability	Consequences of Error	Human Input	Judgement	Outcome Verifiability
1	Navigation	Tactical decision making	Tactical decision making	Tactical decision making	Navigation
2	Maintain flight	Threat assessment	Threat assessment	Threat assessment	Maintain flight
3	Execution of decision	Execution of decision	Maintain flight	Maintain flight	Execution of decision
4	Threat assessment	Maintain flight	Execution of decision	Execution of decision	Threat assessment
5	Tactical decision making	Navigation	Navigation	Navigation	Tactical decision making

From **Table 2** above, we can actually see that navigation most closely fulfils the criteria. This suggests that fully automating this process is the most feasible. Navigation is then followed closely by maintaining flight, with my comparisons offering evidence that tactical decision making and threat assessment are the two roles where full automation is the least likely to happen. This framework supports my hypothesis that AI will first replace humans as the aircraft's operator before the tactical decision maker. It is important to remember that my analysis does not support the idea that the latter will never be fully automated, rather it lays the foundations for the argument that completely removing the human from the loop in these areas will simply take longer.

However, we must also consider the possibility that software-based autonomy actually develops quicker than full physical autonomy, resulting in AI being capable of replacing humans as the tactical decision maker before doing so as the controller. Keeping in mind that the main issue is really if intelligent systems will actually be given the authority to fully make these tactical decisions themselves, there could be a scenario where the AI is analysing the current situation, creating a list of possible actions, and selecting one while a human operator is adding little value. In such a case, the human crew member is still technically 'in-the-loop', but functional replacement may have already happened before full removal of the human pilot.

That being said, comparatively analysing whether flight control or tactical decision making will be replaced by AI first, only works if replacement is defined. In my opinion, the term refers to AI completely

carrying out a role, requiring no human oversight or intervention. With this in mind, the presence of supervision during strategic decision making delays the replacement of the human pilot.

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6. Conclusion

In conclusion, I think that the fighter pilot of the future may no longer be human, but rather AI. My analysis suggests that this transformation, however, will not happen uniformly. Functions such as maintaining flight and navigation are more structured, measurable, and have set aims. On the other hand, threat assessment and decision making require more judgement, are less predictable, and do not yet have the clearance to be fully automated. This supports my hypothesis that intelligent systems will replace humans as the aircraft's controller before the tactical authority, with the latter being a longer process as opposed to being fully ruled out. Ultimately, I believe that human fighter pilots will not disappear in a single step, but instead through a gradual transfer of individual roles.

Word Count = 2,490

7. Evaluation

Most of my projects in the past have been physical artefacts, focusing mainly on their hardware aspects. As such, for my Senior Scholarship, I knew I wanted to try something new and write a research paper rather than make something creative. I have been interested in the military for quite a while, and given that my Arkwright Engineering Scholarship is sponsored by the Royal Air Force, I knew I wanted to do some engineering research in the military field to tie my interests together. Before this project, I had never really done any in-depth study of AI, but more recently have found myself wanting to learn more about it as the use of this technology in robotics is a field I find extremely interesting. Due to the 2500 word limit, I wanted to choose a specific topic within AI in military aircraft. This led me to deciding on the question 'Could the Fighter Pilot of the Future Still be Human?', as it is one that can be tackled using both objective research as well as my own unique analysis.

My main aims for this paper were the following:

1. Gain a deeper understanding of AI
2. Use this to form my own original conclusion about what the fighter pilot of the future looks like
3. Clearly and concisely explain the current implementation of AI within military aircraft
4. Accumulate research from academic papers to inform my own matrix with which to critically analyse the given information

Looking back at the paper, I think I have effectively achieved these goals. Through the course of this project, I have learnt so much about artificial intelligence, and I think I have been able to use my own understanding to communicate my ideas and analysis clearly. Sections 3 and 4 in the paper also set up the knowledge readers need to form their own conclusions about the future of fighter pilots. Going from a question, to looking into existing literature about the field, to then creating my own analysis to find an answer has taught me about the process of research, and has given me clarity in what previously seemed like a convoluted field.

When I started researching and reading papers on AI within military aircraft, I began to second guess how I would find an answer to the question I posed. There were so many factors to take into consideration with fighter pilots, but as I started looking back at the notes I took, the two sides of their job (controlling the aircraft and making tactical decisions), started to become clear. This gave structure to my project, as I then knew what to focus on.

One of the main challenges I faced was the difficulty in getting peer reviewed papers on military technology, as these often required credentials that I did not have. This limited the information I could collate, which meant I had to rely on a combination of academic papers and publicly available information on defense organisation websites. Transitioning from research to writing the paper was also quite hard, especially grouping my notes, and trying to create links between different ideas discovered in other papers. This led to quite a lot of waffle in the first draft of the paper, as it was around 4000 words long! While writing the paper, trying to compare different technologies with varying levels of maturity seemed quite daunting, but using the criteria I established allowed me to break up this comparison into separate aspects.

Although my paper does meet the objectives I set out to achieve, there are a few limitations to my work. My conclusion about the future of fighter pilots is drawn from the criteria that I made, and so could have been different had alternate criteria been chosen. This ties into the fact that part of my analysis is of course subjective, as there is no way to find a concrete answer so relatively early on in the development of AI systems. Building on this, all the criteria are weighted the same, so thus different weights would also change the conclusions produced.

Further development of my project would see a deeper exploration into human-AI teaming and relationship within military aircraft, especially the angle of trust and how they can both collaborate with each other. I would also love to compare my findings with expert analysis and then expand my research into AI use within the Army and Navy, as they would both provide unique insights into the future of defence.

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